

BEE 4750 Homework 4: Linear Programming and Capacity Expansion

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Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `~/BEE 4750/hw4-parker-regina`

```
using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

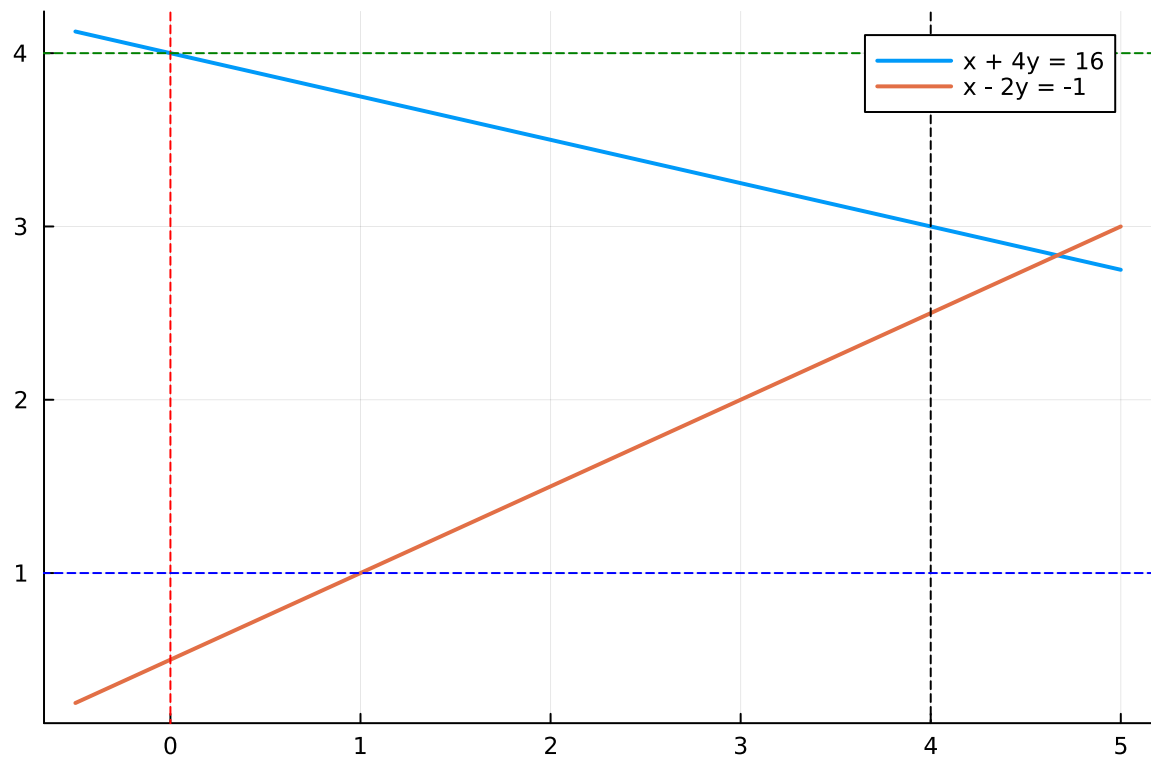
$$\begin{aligned} \max_{x,y} \quad & 3x + 2y \\ \text{subject to:} \quad & x + 4y \leq 16 \\ & x - 2y \leq -1 \\ & 0 \leq x \leq 4 \\ & 1 \leq y \leq 4. \end{aligned}$$

```
#Set x range for plotting
x = -0.5:0.01:5.0

#Constraint functions
y1(x) = (16 - x) / 4
y2(x) = (x + 1) / 2

#Plot constraint functions across all x values
plt = plot(x, y1.(x), lw=2, label="x + 4y = 16")
plot!(plt, x, y2.(x), lw=2, label="x - 2y = -1")

#Draw the vertical and horizontal lines representing
#the bounds for constraints
vline!([0], linestyle=:dash, color=:red, label="")
vline!([4], linestyle=:dash, color=:black, label="")
hline!([1], linestyle=:dash, color=:blue, label="")
hline!([4], linestyle=:dash, color=:green, label="")
```



Using the above graph of constraints, we know that the maximum value must occur at an intersection of the boundaries. We can use these intersections to check the values for the maximum of the function $3x + 2y$. The feasible intersections, as determined by are as follows:

$$(0, 1), (0, 4), (1, 1), (4, \frac{5}{2}), (4, 3)$$

We can then check these pairs with the function $3x + 2y$:

$$3(0) + 2(1) = 2, \quad (0, 1)$$

$$3(0) + 2(4) = 8, \quad (0, 4)$$

$$3(1) + 2(1) = 5, \quad (1, 1)$$

$$3(4) + 2(2.5) = 17, \quad (4, 2.5)$$

$$3(4) + 2(3) = 18, \quad (4, 3)$$

Therefore, we can see that at $(4, 3)$ we achieve the maximum value of $3x + 2y = 18$.

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs \$70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost (\$/ha-yr)	Selling Price (\$/kg)
Soybeans	350	0.36
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear.**

SOLUTION 2.1 To formulate a linear program for this resource allocation problem, we started off by defining nine decision variables: three for each crop of the crops representing the number of hectares allocated to the pesticide application rate of each crop.

$$s_r, w_r, c_r \text{ for } r = 0, 1, 2$$

Each of these decision variables satisfy the non-negativity constraint:

$$s_r, w_r, c_r \geq 0 \text{ for } r = 0, 1, 2$$

The goal is to maximize profit so the objective function follows the equation: \$ profit = revenue - cost \$.

$$\max Z = \sum_{r=0}^2 (p_s y_{s,r} s_r + p_w y_{w,r} w_r + p_c y_{c,r} c_r) - \left(C_s \sum_{r=0}^2 s_r + C_w \sum_{r=0}^2 w_r + C_c \sum_{r=0}^2 c_r \right) - 70 \sum_{r=0}^2 (r s_r + r w_r + r c_r)$$

There are a few constraints included in this LP. First of all, the farm only has 130 ha so the total amount of land used has to be less than or equal to 130 ha.

$$\sum_{r=0}^2 s_r + \sum_{r=0}^2 w_r + \sum_{r=0}^2 c_r \leq 130$$

Second, each crop has a demand of 250,000kg and the production of each crop is capped at this demand limit.

$$\sum_{r=0}^2 y_{s,r} s_r \leq 250000$$

$$\sum_{r=0}^2 y_{w,r} w_r \leq 250000$$

$$\sum_{r=0}^2 y_{c,r} c_r \leq 250000$$

Finally, each crop there is a regulatory limit for each crop on the average pesticide application rate. To find the average pesticide application rate, we used the following equations:

$$\frac{0s_0 + 1s_1 + 2s_2}{s_0 + s_1 + s_2} \leq 0.8$$

$$\frac{0w_0 + 1w_1 + 2w_2}{w_0 + w_1 + w_2} \leq 0.7$$

$$\frac{0c_0 + 1c_1 + 2c_2}{c_0 + c_1 + c_2} \leq 0.6$$

However, all the constraints must be linear so the equations were manipulated to:

$$0s_0 + 1s_1 + 2s_2 \leq 0.8(s_0 + s_1 + s_2)$$

$$0w_0 + 1w_1 + 2w_2 \leq 0.7(w_0 + w_1 + w_2)$$

$$0c_0 + 1c_1 + 2c_2 \leq 0.6(c_0 + c_1 + c_2)$$

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

SOLUTION 2.2 The linear program recommends:

Soybeans: 69.06 ha total

13.81 ha at 0 kg/ha

55.25 ha at 1 kg/ha

Wheat: 22.48 ha total

6.74 ha at 0 kg/ha

15.73 ha at 1 kg/ha

Corn: 38.46 ha total

26.92 ha at 0 kg/ha

11.54 ha at 2 kg/ha

This will yield a maximized profit of \$ 116,741 while remaining within the land constraint of 130 ha limit.

To summarize, to maximize profit the farmer should plant approximately 69.06 ha of soybeans primarily using application rate 1, 22.48 ha of wheat also mostly using application rate 1, and 38.46 ha of corn using a mix of rate 0 and rate 2 applications.

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

The shadow price of the land constraint is \$ 729,40 /ha and this represents the increase in profit if crops were grown on one additional hectare of land.

Problem 2 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs \$70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

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Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear.**

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Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

SOLUTION 2.3

The shadow price of the land constraint is \$ 729.40 / ha. This represents the increase in profit if one additional hectare of land were available to grow crops on.

So, the value of an additional 10 hectares of land is:

$$\$ 10 * 729.40 = \$ 7,294 \$$$

According to the results from the LP, soybeans and corn have both hit the 250,000 kg demand limit. This means that the farmer could only plan wheat on the 10 extra ha of land since that is the only crop that has not hit the demand limit. From our findings explained in 2.2 we know that for wheat we will apply $\frac{6.74}{22.48} * 100 = 30\%$ at rate 0 and $\frac{15.74}{22.48} * 100 = 70\%$ at rate 1 and so the yield is $\$ 0.3 * 3500 + 0.7 * 4100 = 3920\$$ kg/ha. Hence, the revenue is $0.27 * 3920 = \$ 1,058.40$ / ha, the production cost is \$ 280 / ha, and the pesticide cost is $70 * 0.7 = \$ 49$ / ha. This results in the profit per extra ha to be:

$$1,058.40 - 280 - 49 = \$ 729.40 / \text{extra ha}$$

This is exactly the shadow price of land so this value of \$ 7,294 makes sense!

The farmer should make the purchase if the annualized cost of buying 10 hectares is less than \$ 7,294.

Problem 2 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

```
#Problem 2.1
using JuMP # optimization modeling syntax
using HiGHS # optimization solver

crop_model = Model(HiGHS.Optimizer) # initialize model object
@variable(crop_model, s[0:2] >= 0) # non-neg constraints for soy
@variable(crop_model, w[0:2] >= 0) # non-neg constraints for wheat
@variable(crop_model, c[0:2] >= 0) # non-neg constraints for corn

# index: 1 = soy, 2 = wheat, 3 = corn
```

```

# set up variables:
# selling prices ($/kg)
ps, pw, pc = 0.36, 0.27, 0.22
# production cost ($/ha)
cs, cw, cc = 350, 280, 390
# pesticide cost ($ per kg/ha applied)
pest = 70
# yields (kg/ha)
ys0, ys1, ys2 = 2900, 3800, 4400
yw0, yw1, yw2 = 3500, 4100, 4200
yc0, yc1, yc2 = 5900, 6700, 7900
# average rate caps (kg/ha)
Ls, Lw, Lc = 0.8, 0.7, 0.6
# land & demand
land_cap = 130
demand = 250000 #for each crop

# OBJ: maximize profits over 130 total ha while remaining in regulatory
# compliance if demand for each crop (which is the maximum the market
# would buy) this year is 250,000 kg
# Objective = revenue - production costs - pesticide costs
@objective(crop_model, Max,
    # revenue
    + ps*(ys0*s[0] + ys1*s[1] + ys2*s[2])
    + pw*(yw0*w[0] + yw1*w[1] + yw2*w[2])
    + pc*(yc0*c[0] + yc1*c[1] + yc2*c[2])
    # production costs
    - cs*sum(s[i] for i in 0:2)
    - cw*sum(w[i] for i in 0:2)
    - cc*sum(c[i] for i in 0:2)
    # pesticide costs (rate * pest * hectares)
    - pest*( 0*s[0] + 1*s[1] + 2*s[2]
            + 0*w[0] + 1*w[1] + 2*w[2]
            + 0*c[0] + 1*c[1] + 2*c[2])
)

# Constraints

# Land
@constraint(crop_model, land, sum(s[i] for i in 0:2) + sum(w[i] for i in 0:2) + sum(c[i] for
# Demand caps (kg)

```

```

@constraint(crop_model, ds, ys0*s[0] + ys1*s[1] + ys2*s[2] <= demand) # soy
@constraint(crop_model, dw, yw0*w[0] + yw1*w[1] + yw2*w[2] <= demand) # wheat
@constraint(crop_model, dc, yc0*c[0] + yc1*c[1] + yc2*c[2] <= demand) # corn

# Average pesticide caps: sum(rate*x) L * sum(x)
@constraint(crop_model, pest_s, (0*s[0] + 1*s[1] + 2*s[2]) <= Ls*(sum(s[i] for i in 0:2)))
@constraint(crop_model, pest_w, (0*w[0] + 1*w[1] + 2*w[2]) <= Lw*(sum(w[i] for i in 0:2)))
@constraint(crop_model, pest_c, (0*c[0] + 1*c[1] + 2*c[2]) <= Lc*(sum(c[i] for i in 0:2)))

print(crop_model)

```

Max 694 s[0] + 948 s[1] + 1094 s[2] + 665.00000000000001 w[0] + 757 w[1] + 714 w[2] + 908 c[0]

Subject to

```

land : s[0] + s[1] + s[2] + w[0] + w[1] + w[2] + c[0] + c[1] + c[2] 130
ds : 2900 s[0] + 3800 s[1] + 4400 s[2] 250000
dw : 3500 w[0] + 4100 w[1] + 4200 w[2] 250000
dc : 5900 c[0] + 6700 c[1] + 7900 c[2] 250000
pest_s : -0.8 s[0] + 0.1999999999999996 s[1] + 1.2 s[2] 0
pest_w : -0.7 w[0] + 0.30000000000000004 w[1] + 1.3 w[2] 0
pest_c : -0.6 c[0] + 0.4 c[1] + 1.4 c[2] 0
s[0] 0
s[1] 0
s[2] 0
w[0] 0
w[1] 0
w[2] 0
c[0] 0
c[1] 0
c[2] 0

```

```

optimize!(crop_model);

println("Optimized soybean values:")
@show value.(s[i] for i in 0:2);

println("Optimized wheat values:")
@show value.(w[i] for i in 0:2);

println("Optimized corn values:")
@show value.(c[i] for i in 0:2);

```

```
println("Optimized objective value:")
@show objective_value(crop_model);
```

LP has 7 rows; 9 cols; 27 nonzeros

Coefficient ranges:

Matrix [2e-01, 8e+03]

Cost [7e+02, 1e+03]

Bound [0e+00, 0e+00]

RHS [1e+02, 2e+05]

Solving LP without presolve, or with basis, or unconstrained

Model status : Optimal

Objective value : 1.1674116702e+05

P-D objective error : 0.0000000000e+00

HiGHS run time : 0.00

Optimized soybean values:

value.((s[i] for i = 0:2)) = [13.812154696132604, 55.24861878453038, 0.0]

Optimized wheat values:

value.((w[i] for i = 0:2)) = [6.743306417339563, 15.734381640458977, 0.0]

Optimized corn values:

value.((c[i] for i = 0:2)) = [26.92307692307692, 0.0, 11.538461538461547]

Optimized objective value:

objective_value(crop_model) = 116741.16702082448

```
println("Shadow values:")
@show shadow_price(land);
```

```
@show shadow_price(ds);
```

```
@show shadow_price(dw);
```

```
@show shadow_price(dc);
```

```
@show shadow_price(pest_w);
```

```
@show shadow_price(pest_w);
```

```
@show shadow_price(pest_c);
```

Shadow values:

shadow_price(land) = 729.4

shadow_price(ds) = 0.046353591160220996

shadow_price(dw) = -0.0

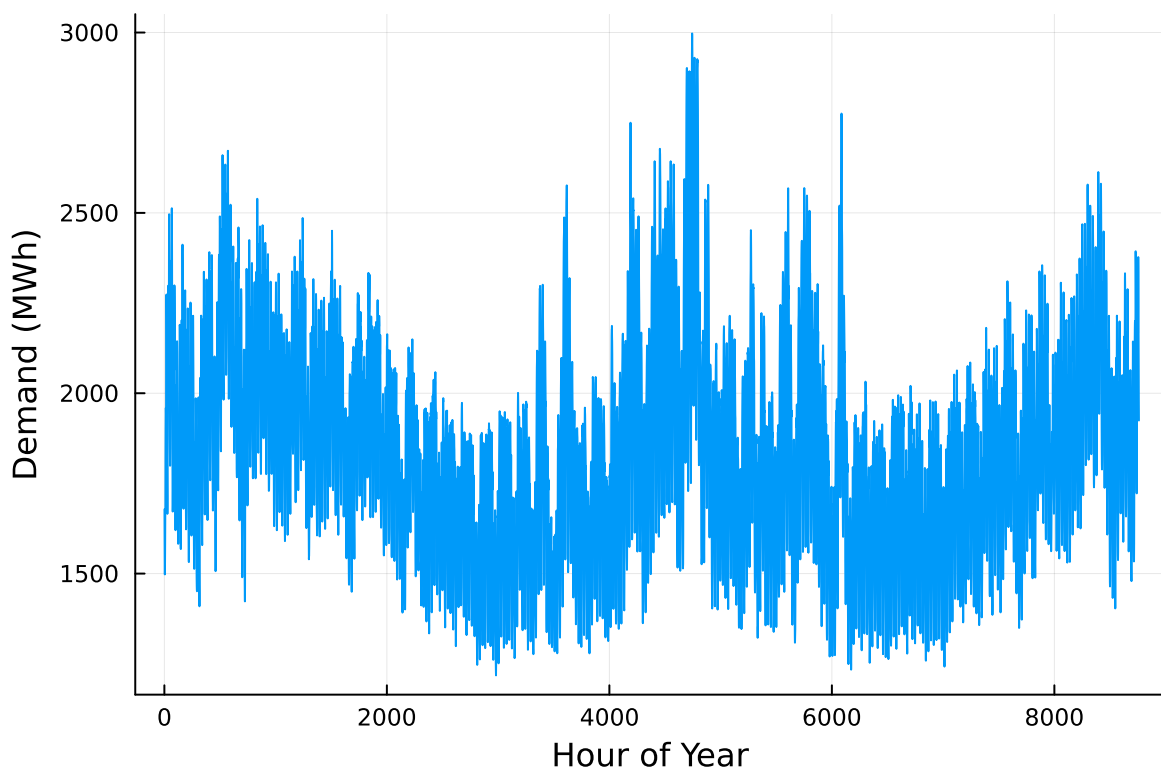
shadow_price(dc) = 0.04132307692307693

shadow_price(pest_w) = 91.99999999999993

```
shadow_price(pest_w) = 91.99999999999993
shadow_price(pest_c) = 108.67692307692309
```

```
# load the data, pull Zone C, and reformat the DataFrame
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))
rename!(NY_demand, :Time Stamp => :Date)
demand = NY_demand[:, [:Date, :C]]
rename!(demand, :C => :Demand)
demand[:, :Hour] = 1:nrow(demand)

# plot demand
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)
```



Next, we load the generator data, shown in Table 3. This data includes fixed costs (\$/MW installed), variable costs (\$/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```

	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

```
# load capacity factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

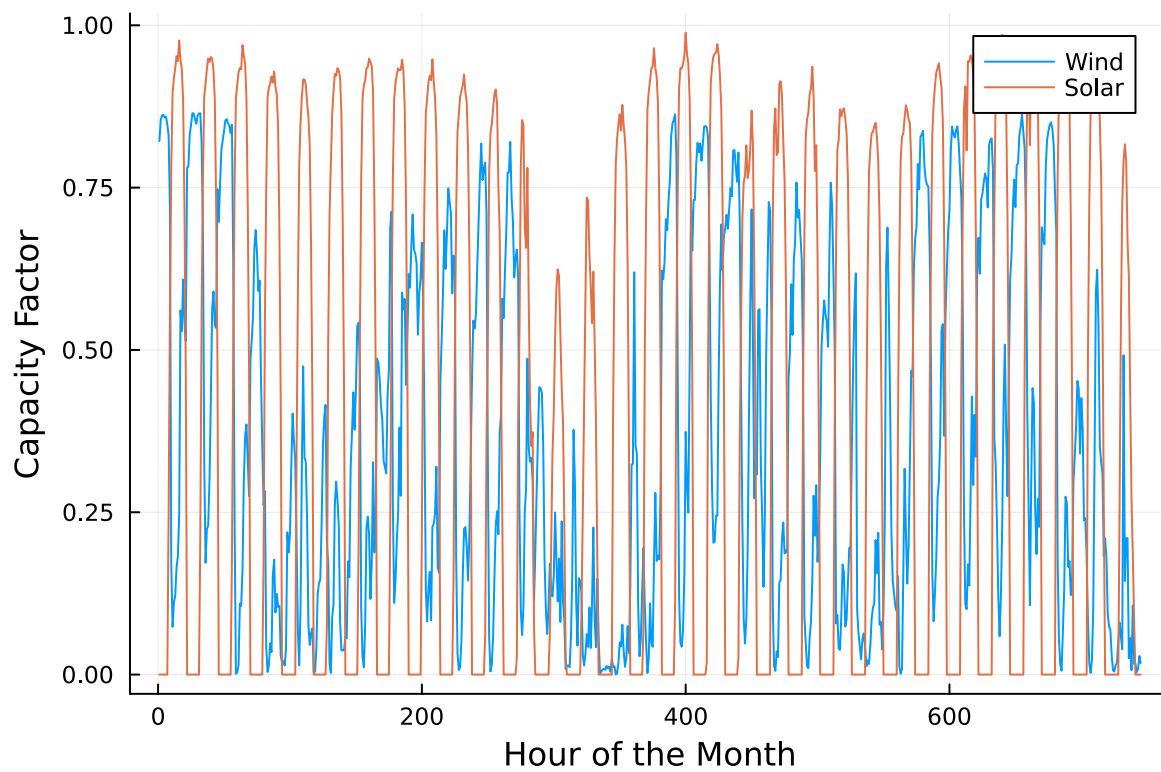
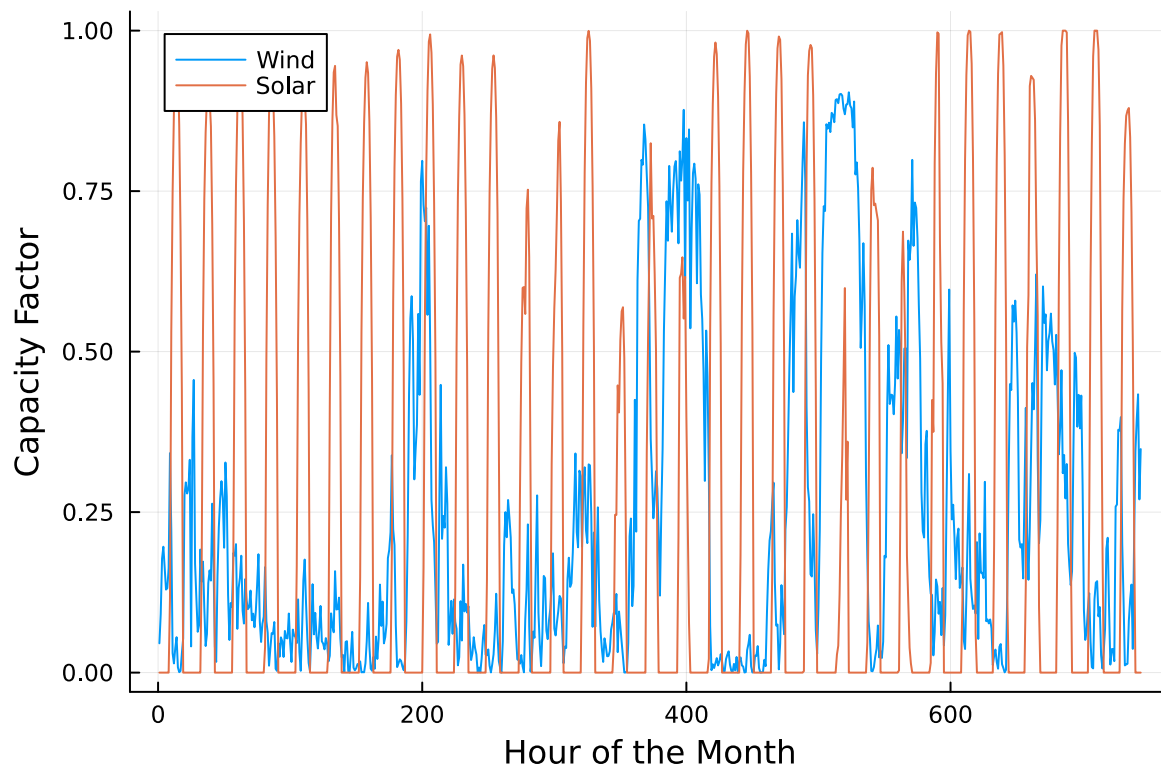
# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
xaxis!("Hour of the Month")
yaxis!("Capacity Factor")

display(p1)
display(p2)
```

You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO₂ limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO₂/yr.

While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of \$10,000/MWh.



Problem 3.1

Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

$$\begin{aligned}
& \min_{x,y,NSE} \quad \sum_{g \in \mathcal{G}} \text{FixedCost}_g \times x_g + \sum_{t \in \mathcal{T}} \sum_{g \in \mathcal{G}} \text{VarCost}_g \times y_{g,t} + \sum_{t \in \mathcal{T}} \text{NSECost} \times NSE_t \\
& \text{subject to:} \quad \sum_{g \in \mathcal{G}} y_{g,t} + NSE_t \geq d_t & \forall t \in \mathcal{T} \\
& \quad y_{g,t} \leq \eta_{g,t} \times x_g & \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \\
& \quad \sum_{t \in \mathcal{T}} \sum_{g \in \mathcal{G}} \text{Emissions}_g \times y_{g,t} \leq TE & \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \\
& \quad x_g, y_{g,t}, NSE_t \geq 0 & \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \\
& & (1)
\end{aligned}$$

** Note these equations are the same as used in lecture except for Emissions_g and $\eta_{g,t}$

Decision Variables:

x_g : Installed capacity of generation technology g [MW]

$y_{g,t}$: Energy generated by technology g during time t [MWh]

NSE_t : Non-served energy during time period t [MWh]

Parameters:

FixedCost_g : Annualized fixed cost per MW of installed capacity for technology g [\$/MW-yr]

VarCost_g : Variable operating cost of technology g [\$/MWh]

NSECost : Penalty cost for non-served energy [\$/MWh] = 10,000 \$/MWh

d_t : Electricity demand over time t [MWh].

$\eta_{g,t}$: Capacity factor of technology g in time period t

Coal, CCGT, CT : $\eta_{g,t} = 1$

Geothermal : $\eta_{g,t} = 0.85$

Solar, Wind : $\eta_{g,t}$ varies hourly based on meteorology

Emissions_g : CO_2 emissions rate of technology g [tCO₂/MWh]

TE : Annual CO_2 emissions limit = 1.5 MtCO₂/yr

$\mathcal{G}$: Set of generation technologies: {coal, CCGT, CT, geothermal, solar, wind}

$\mathcal{T}$: Total time scale

Plant [g]	FixedCost [\$]	VarCost [\$/MWh]	Emissions [tCO ₂ /MWh]
Geothermal	450000	0	0.00
Coal	220000	24	1.00
NG CCGT	82000	30	0.43
NG CT	65000	40	0.55
Wind	91000	0	0.00
Solar	70000	0	0.00

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?

Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

Problem 3.4

Make a plot of the electricity price in each hour. Discuss any trends that you see.

Problem 3.5

What is the shadow price of CO₂? What is the value to the utility be of allowing it to emit an additional 1000 tCO₂/yr?

Significant Digits

Use `round(x; digits=n)` to report values to the appropriate precision! If your number is on a different order of magnitude and you want to round to a certain number of significant digits, you can use `round(x; sigdigits=n)`.

Getting Variable Output Values

`value.(x)` will report the values of a JuMP variable `x`, but it will return a special container which holds other information about `x` that is useful for JuMP. This means that you can't use this output directly for further calculations. To just extract the values, use `value.(x).data`.

Suppressing Model Command Output

The output of specifying model components (variable or constraints) can be quite large for this problem because of the number of time periods. If you end a cell with an `@variable` or `@constraint` command, I *highly* recommend suppressing output by adding a semi-colon after the last command, or you might find that your notebook crashes.

```
# define sets
G = 1:nrow(gens[1:end-2, :])
T = 1:nrow(demand)
NSECost = 10000

gencap = Model(HiGHS.Optimizer)
# define variables
@variables(gencap, begin
    x[g in G] >= 0
    y[g in G, t in T] >= 0
    NSE[t in T] >= 0
end)
@objective(gencap, Min,
    sum(gens[G, :FixedCost] .* x) + sum(gens[G, :VarCost] .* sum(y[:, t] for t in T)) + NSECost
)
@constraint(gencap, load[t in T], sum(y[:, t]) + NSE[t] >= demand.Demand[t])
@constraint(gencap, availability[g in G, t in T], y[g, t] <= x[g])
optimize!(gencap)
```

Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms
LP has 43800 rows; 43804 cols; 113880 nonzeros

Coefficient ranges:

Matrix	[1e+00, 1e+00]
Cost	[2e+01, 4e+05]
Bound	[0e+00, 0e+00]
RHS	[1e+03, 3e+03]

Presolving model

43800 rows, 43804 cols, 113880 nonzeros 0s

43799 rows, 43803 cols, 113878 nonzeros 0s

Presolve : Reductions: rows 43799(-1); columns 43803(-1); elements 113878(-2)

Solving the presolved LP

Using EKK dual simplex solver - serial

Iteration	Objective	Infeasibilities	num(sum)
0	0.0000000000e+00	Pr: 8760(1.63679e+07)	0s
27989	7.2267190900e+08	Pr: 0(0)	0s

```
Solving the original LP from the solution after postsolve
Model status      : Optimal
Simplex iterations: 27989
Objective value    : 7.2267190900e+08
P-D objective error : 0.0000000000e+00
HiGHS run time     : 0.25
```

References

List any external references consulted, including classmates.